

Ed Dubnitsky

ML Engineer · Backend & Data · 12+ years

Fullstack engineer with 12+ years of shipping software, the past year fully on production AI. Comfortable with training and applying neural-network models (PyTorch, Transformers), computer vision (OpenCV), NLP, and LLM integration. Strong background in data visualisation and SQL analytics.

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Remote · EU-aligned working hours · English C1 · Russian native

Target Role

ML Engineer — production RAG, computer vision, multi-provider LLM pipelines.

Full-time, remote. Open to relocation and business travel.

Core Competencies

ML / AI: PyTorch, TensorFlow, Transformers (HuggingFace), OpenCV, NLP, Computer Vision, LLM integration (GPT-4, Claude, Ollama, Groq, NVIDIA NIM), RAG, Qdrant, embeddings, semantic search

Backend: Python, FastAPI, asyncio, Pydantic, Celery, Flask, Django, Node.js / Express, REST, WebSocket, SSE

Data: NumPy, Pandas, SciPy, Matplotlib, Plotly, SQL analytics, ECharts, Dremio

Databases: PostgreSQL, SQL Server (T-SQL, SPXML), Redis, MongoDB

Frontend: React, Vue.js, TypeScript, Next.js, Nuxt, Astro, HTML5 / CSS3

Infrastructure: Docker, Docker Compose, Ubuntu / Linux, nginx, CI/CD, GitHub Actions, Prometheus, Git

Quality: pytest, mypy, ruff, evaluation harnesses, ADRs, structured logging

Other: Telegram Bot API, REST API design, WebSocket, FFmpeg

Experience — 12+ years

ML Engineer — Go-Active (Telegram Mini App)

April 2025 — present · Remote

GPS-route visualization and analysis platform with ML analytics. Built the Telegram Mini App from scratch: GPS-track upload, cinematic 3D-video generation, computer-vision module for render-quality scoring and camera-trajectory reconstruction from video (Structure-from-Motion).

- Implemented the ML pipeline: depth estimation (MiDaS / PyTorch), 15+ video-quality metrics, automatic route-pattern classification
- Designed the service architecture: 5-stage render pipeline, memory monitoring, Docker container orchestration
- Test coverage with pytest, strict typing with mypy, linting with ruff

Stack: Python, PyTorch, OpenCV, NumPy / SciPy / Pandas, Pydantic, React, TypeScript, Docker, FFmpeg, CesiumJS

Frontend Engineer — MCCME (Moscow Centre for Continuous Mathematical Education)

June 2023 — March 2025 (1 yr 10 mo) · Moscow · 7.math.ru

- Built a course-editing platform for math curricula from scratch (React, TypeScript, Redux Toolkit, Redux-Saga)
- Rewrote the publication admin panel from scratch
- Component refactor and React upgrade to current version

Analytics Engineer — GoodData

September 2022 — May 2023 (9 mo) · Prague, Czech Republic · gooddata.com

- Built data marts with interactive charts for the BI low-code platform from scratch
- Customizable data-visualization widgets: metrics, dimensions, thresholds, gradients
- Worked in a small team with the analytics lead; code review in GitLab

Stack: Vue 2, ECharts, TypeScript, Dremio, Git

Lead Analytics Engineer — MediaMarktSaturn (L&D Platform)

August 2018 — July 2022 (4 yrs) · Ingolstadt, Germany · mediamarktsaturn.com

Internal learning & development analytics platform for tens of thousands of store associates across the retail group.

- Built and ran the L&D analytics platform (frontend + backend). Promoted to tech lead after the first two years, ran the department solo for ~6 months and rebuilt the team from scratch
- Designed the T-SQL / SPXML reporting layer: complex threshold- and gradient-driven queries with parameterised triggers
- Migrated the UI to Vuetify; built new manager and employee dashboards
- Cut bundle payload by 50%
- Authored SQL endpoints for every new interface

Stack: Vue.js, Nuxt, Vuetify, JavaScript ES6, SQL Server (T-SQL, SPXML), Sass, Vite

Web Developer — 101XP

May 2017 — March 2018 (11 mo) · Moscow · 101xp.com

- Cross-browser, responsive markup for promo pages and site sections
- React, Node.js, Docker, Python / Django

Webmaster — iruntheflight.com (Independent)

September 2014 — May 2017 (2 yrs 9 mo) · Remote

- Built and ran a specialised aviation web service
- Picked up Python, Flask, Jekyll

Systems Administrator — NP GLONASS

February 2013 — September 2014 (1 yr 8 mo) · Moscow · glonassunion.ru

- Backup rollout (Symantec), virtualisation (VMware ESX)
- Linux, Mikrotik, PostgreSQL, nginx, Tomcat administration
- 24/7 IT-service monitoring

Personal Projects

Astro-Scheduler — astronomical scheduling with ML

Fullstack platform for personalised scheduling: 17+ astronomical data layers, LLM pipeline for rule generation, constraint-satisfaction solver for schedule optimisation.

- Python / FastAPI service for astronomical computations (ephemerides, lunar phases, transits)
- ML feedback loop: the system learns from outcomes and adjusts recommendation weights
- GPT-4 / Claude integration for natural-language rule generation and validation
- PostgreSQL for natal charts, rules, user feedback

Stack: Python, FastAPI, PostgreSQL, React, TypeScript, Docker, OpenAI API, NumPy

YouTube Voice Actor — AI system for analysing and voicing YouTube comments (production)

Telegram bot and web app that parses YouTube comments, runs emotion analysis, and synthesises speech with voice cloning.

- NLP pipeline: language detection (16 languages), emotion analysis (RoBERTa / Transformers), sentiment classification
- Neural TTS (XTTS v2) with voice cloning and emotional colouring
- Two interfaces: Telegram bot (python-telegram-bot) and web app (FastAPI + frontend)

Stack: Python, FastAPI, Flask, PyTorch, Transformers, XTTS, Telegram Bot API

OpenClaw AI Assistant (Cronos)

Personal AI assistant on the OpenClaw platform for automating job search, task management, and service integration.

- Built skills for vacancy parsing across Telegram channels and job boards; automated morning and evening parsers
- Heartbeat system (HEARTBEAT.md) for periodic email, calendar, and vacancy checks without manual intervention
- Integrated exec, memory, and filesystem tools for reading and analysing CVs, vacancies, and configs
- Safe-mode operation: mandatory command approval via exec
- Markdown knowledge base (MEMORY.md, dated memory) for long-term context and learning from past sessions

Stack: Python, OpenClaw CLI, Telegram API, Git, Markdown, Bash

Education

Moscow State Open University (V.S. Chernomyrdin) — 2008

Applied Mathematics and Informatics · Major: Global Information Management Systems

Continued education: HTML Academy — JavaScript Developer, advanced course (2018)

Languages & Other

Languages: Russian — native; English — C1 (fluent spoken and written)

Driving licence: A, B